

Лабораторная работа №5

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

2026-02-16

1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Приобретение практических навыков расширенного конфигурирования HTTP-сервера Apache

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- Настройка безопасности (HTTPS)

1.1 Цель работы

- Приобретение практических навыков расширенного конфигурирования HTTP-сервера Apache
- Настройка безопасности (HTTPS)
- Интеграция поддержки PHP

Раздел 2

2. Выполнение лабораторной работы

2.1 Конфигурирование HTTP-сервера для работы через протокол HTTPS

- В каталоге `/etc/pki/tls/private` создан закрытый ключ и самоподписанный сертификат
- Использована команда `openssl req -x509 -nodes -newkey rsa:2048`

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# mkdir -p /etc/pki/tls/private
[root@server.bahis.net ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server.bahis.net ~]# cd /etc/pki/tls/private
[root@server.bahis.net private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.
ser.net.key -out www.user.net.crt
.....+.....+.....+*****x.....+*****+
*****+.....+.....+*****
...+...+...+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
...+...+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
*****+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
...+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
```

2.1 Конфигурирование HTTP-сервера для работы через протокол HTTPS

- В каталоге `/etc/pki/tls/private` создан закрытый ключ и самоподписанный сертификат
- Использована команда `openssl req -x509 -nodes -newkey rsa:2048`
- Заданы параметры DN (RU, Russia, Moscow, bahis, CN=bahis.net)

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# mkdir -p /etc/pki/tls/private
[root@server.bahis.net ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server.bahis.net ~]# cd /etc/pki/tls/private
[root@server.bahis.net private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.
ser.net.key -out www.user.net.crt
.....+.....+.....+*****x.....+*****+
*****+.....+.....+*****
...+...+...+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
...+...+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
*****+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
...+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+.....+
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- Использована команда `openssl req -x509 -nodes -newkey rsa:2048`
- Заданы параметры DN (RU, Russia, Moscow, bahis, CN=bahis.net)
- Сертификат скопирован в `/etc/ssl/certs`

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# mkdir -p /etc/pki/tls/private
[root@server.bahis.net ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server.bahis.net ~]# cd /etc/pki/tls/private
[root@server.bahis.net private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.
ser.net.key -out www.user.net.crt
.....+.....+.....+*****+.....+*****+
*****+.....+.....+*****+
...+...+...+.....+*****+.....+...+...+...+.....+
...+...+.....+...+.....+...+...+...+...+...+...+*****+
*****+.....+...+...+...+...+...+...+...+...+...+
...+...+...+.....+
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You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
```

- Отредактирован файл `www.bahis.net.conf`

```
root@server:/etc/httpd/conf.d
GNU nano 5.6.1 /etc/httpd/conf.d/www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
```

- Отредактирован файл `www.bahis.net.conf`
- В `<VirtualHost *:80>` настроена переадресация на HTTPS (`RewriteRule`)

```
root@server:/etc/httpd/conf.d
GNU nano 5.6.1 /etc/httpd/conf.d/www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
```

- Отредактирован файл `www.bahis.net.conf`
- В `<VirtualHost *:80>` настроена переадресация на HTTPS (`RewriteRule`)
- В `<VirtualHost *:443>` включён `SSLEngine on`

```
root@server:/etc/httpd/conf.d
GNU nano 5.6.1 /etc/httpd/conf.d/www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
```

- Отредактирован файл `www.bahis.net.conf`
- В `<VirtualHost *:80>` настроена переадресация на HTTPS (`RewriteRule`)
- В `<VirtualHost *:443>` включён `SSLEngine on`
- Указаны `SSLCertificateFile` и `SSLCertificateKeyFile`

```
root@server:/etc/httpd/conf.d
GNU nano 5.6.1 /etc/httpd/conf.d/www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
```

- Отредактирован файл `www.bahis.net.conf`
- В `<VirtualHost *:80>` настроена переадресация на HTTPS (`RewriteRule`)
- В `<VirtualHost *:443>` включён `SSLEngine on`
- Указаны `SSLCertificateFile` и `SSLCertificateKeyFile`
- Настроены журналы и `DocumentRoot`

```
root@server:/etc/httpd/conf.d
GNU nano 5.6.1 /etc/httpd/conf.d/www.bahis.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
ErrorLog logs/www.bahis.net-error_log
CustomLog logs/www.bahis.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@bahis.net
DocumentRoot /var/www/html/www.bahis.net
ServerName www.bahis.net
```


- Проверены активные сервисы firewall

```
[root@server.bahis.net conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http ssh
[root@server.bahis.net conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcu
psd audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-stor
age bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph c
eph-exporter ceph-mon cfengine checkmk-agent cockpit collectd condor-collector cratedb c
tdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls dock
er-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fo
reman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-tru
st ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availability http
http3 https ident imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins
s kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-man
ager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kube
let kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llm
nr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mo
sh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboar
d nfs nfs3 nmap nmap-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconso
le ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql proxyo pr
ometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmast
er quassel radius rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-m
aster samba samba-client samba-dc sane sip sips slp smtp smtp-submission smtps snmp snmp
tls snmptls-trap snmptrap spideroak-lansync spotify-sync squid sssd ssh steam-streaming
svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tenta
cle tftp tile38 tinc tor-socks transmission-client upnp-client vdsd vnc-server warpinato
r whem-http whem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-di
```

- Проверены активные сервисы firewallld
- Добавлен сервис https

```
[root@server.bahis.net conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http ssh
[root@server.bahis.net conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcu
psd audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-stor
age bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph c
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tdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls dock
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reman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-tru
st ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availability http
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ometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmast
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r whem-http whem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-di
```

- Проверены активные сервисы firewalld
- Добавлен сервис https
- Правила сохранены как постоянные

```
[root@server.bahis.net conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http ssh
[root@server.bahis.net conf.d]# firewall-cmd --get-services
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tdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-tls dock
er-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fo
reman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-tru
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```

- Проверены активные сервисы firewallld
- Добавлен сервис https
- Правила сохранены как постоянные
- Выполнен перезапуск httpd

```
[root@server.bahis.net conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http ssh
[root@server.bahis.net conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcu
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cle tftp tile38 tinc tor-socks transmission-client unpn-client vdsd vnc-server warpinato
r whem-http whem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-di
```

- На клиенте открыт `https://www.bahis.net`

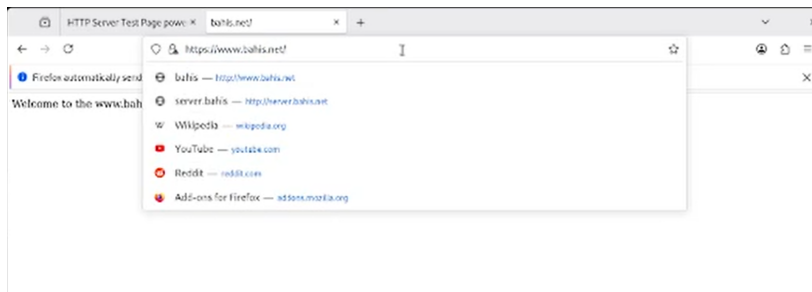


Рисунок 4: Доступ к веб-серверу по HTTPS в браузере клиента

- На клиенте открыт `https://www.bahis.net`
- Выполнено перенаправление HTTP → HTTPS

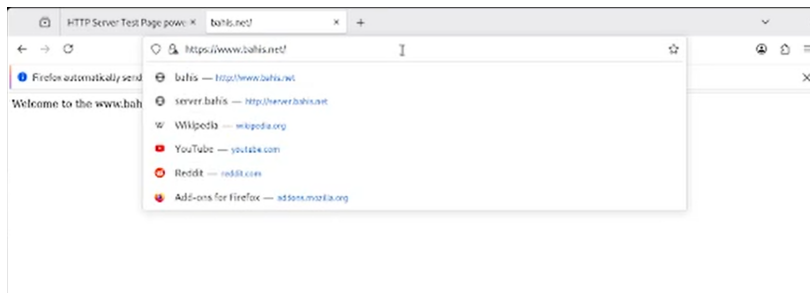


Рисунок 4: Доступ к веб-серверу по HTTPS в браузере клиента

- На клиенте открыт `https://www.bahis.net`
- Выполнено перенаправление HTTP → HTTPS
- После подтверждения исключения отображается страница

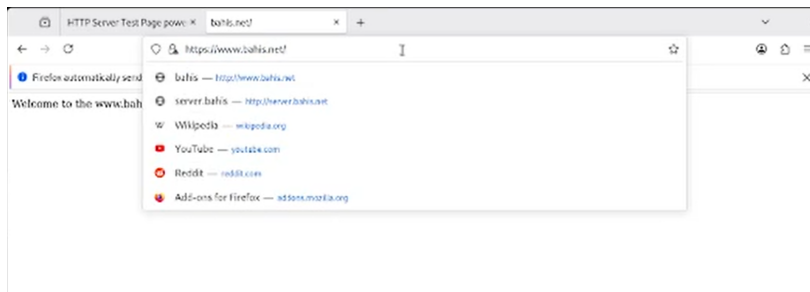


Рисунок 4: Доступ к веб-серверу по HTTPS в браузере клиента

- На клиенте открыт `https://www.bahis.net`
- Выполнено перенаправление HTTP → HTTPS
- После подтверждения исключения отображается страница
- Подтверждена работа HTTPS

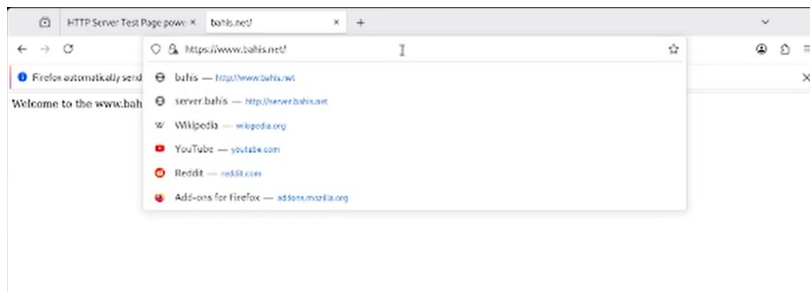


Рисунок 4: Доступ к веб-серверу по HTTPS в браузере клиента

2.2 Конфигурирование HTTP-сервера для работы с PHP

- Выполнена установка пакета php через dnf

```
[root@server.bahis.net ~]# dnf -y install php
Rocky Linux 9 - BaseOS                    5.2 kB/s | 4.3 kB    00:00
Rocky Linux 9 - AppStream                  8.7 kB/s | 4.8 kB    00:00
Rocky Linux 9 - Extras                     5.7 kB/s | 3.1 kB    00:00
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
php                                    x86_64            8.0.30-4.el9_7    appstream         8.6 k
Installing dependencies:
nginxfilesystem                       noarch            2:1.20.1-24.el9   appstream         9.2 k
php-common                             x86_64            8.0.30-4.el9_7    appstream        666 k
Installing weak dependencies:
php-cli                               x86_64            8.0.30-4.el9_7    appstream        3.1 M
php-fpm                               x86_64            8.0.30-4.el9_7    appstream        1.6 M
php-mbstring                           x86_64            8.0.30-4.el9_7    appstream        468 k
php-opcache                             x86_64            8.0.30-4.el9_7    appstream        510 k
php-pdo                                 x86_64            8.0.30-4.el9_7    appstream         81 k
php-xml                                 x86_64            8.0.30-4.el9_7    appstream        132 k

Transaction Summary
=====
Install 9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginxfilesystem-1.20.1-24.el9.noarch.rpm    37 kB/s | 9.2 kB    00:00
```

2.2 Конфигурирование HTTP-сервера для работы с PHP

- Выполнена установка пакета php через dnf
- Установлена версия PHP 8.0.30

```
[root@server.bahis.net ~]# dnf -y install php
Rocky Linux 9 - BaseOS                    5.2 kB/s | 4.3 kB    00:00
Rocky Linux 9 - AppStream                  8.7 kB/s | 4.8 kB    00:00
Rocky Linux 9 - Extras                    5.7 kB/s | 3.1 kB    00:00
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
php                                    x86_64            8.0.30-4.el9_7    appstream         8.6 k
Installing dependencies:
nginxfilesystem                       noarch            2:1.20.1-24.el9   appstream         9.2 k
php-common                             x86_64            8.0.30-4.el9_7    appstream         666 k
Installing weak dependencies:
php-cli                               x86_64            8.0.30-4.el9_7    appstream         3.1 M
php-fpm                               x86_64            8.0.30-4.el9_7    appstream         1.6 M
php-mbstring                          x86_64            8.0.30-4.el9_7    appstream         468 k
php-opcache                           x86_64            8.0.30-4.el9_7    appstream         510 k
php-pdo                               x86_64            8.0.30-4.el9_7    appstream         81 k
php-xml                               x86_64            8.0.30-4.el9_7    appstream         132 k

Transaction Summary
=====
Install 9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginxfilesystem-1.20.1-24.el9.noarch.rpm 37 kB/s | 9.2 kB    00:00
```

2.2 Конфигурирование HTTP-сервера для работы с PHP

- Выполнена установка пакета php через dnf
- Установлена версия PHP 8.0.30
- Установлены необходимые зависимости

```
[root@server.bahis.net ~]# dnf -y install php
Rocky Linux 9 - BaseOS                    5.2 kB/s | 4.3 kB    00:00
Rocky Linux 9 - AppStream                  8.7 kB/s | 4.8 kB    00:00
Rocky Linux 9 - Extras                     5.7 kB/s | 3.1 kB    00:00
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
php                                     x86_64            8.0.30-4.el9_7    appstream          8.6 k
Installing dependencies:
nginxfilesystem                        noarch            2:1.20.1-24.el9   appstream          9.2 k
php-common                             x86_64            8.0.30-4.el9_7    appstream         666 k
Installing weak dependencies:
php-cli                               x86_64            8.0.30-4.el9_7    appstream         3.1 M
php-fpm                               x86_64            8.0.30-4.el9_7    appstream         1.6 M
php-mbstring                          x86_64            8.0.30-4.el9_7    appstream          468 k
php-opcache                           x86_64            8.0.30-4.el9_7    appstream          510 k
php-pdo                               x86_64            8.0.30-4.el9_7    appstream           81 k
php-xml                               x86_64            8.0.30-4.el9_7    appstream          132 k

Transaction Summary
=====
Install 9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginxfilesystem-1.20.1-24.el9.noarch.rpm 37 kB/s | 9.2 kB    00:00
```

- Удалён index.html

```
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
mv: cannot stat '/etc/resolv.conf.bak': No such file or directory
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# cd /var/www/html/www.bahis.net
[root@server.bahis.net www.bahis.net]# ls
index.html
0:17:47 [root@server.bahis.net www.bahis.net]# rm index.html
rm: remove regular file 'index.html'? y
[root@server.bahis.net www.bahis.net]# nano index.php
```

Рисунок 6: Создание index.php, настройка прав доступа и перезапуск httpd

- Удалён index.html
- Создан файл index.php с phpinfo();

```
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
mv: cannot stat '/etc/resolv.conf.bak': No such file or directory
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# cd /var/www/html/www.bahis.net
[root@server.bahis.net www.bahis.net]# ls
index.html
0:17:47 [root@server.bahis.net www.bahis.net]# rm index.html
rm: remove regular file 'index.html'? y
[root@server.bahis.net www.bahis.net]# nano index.php
```

Рисунок 6: Создание index.php, настройка прав доступа и перезапуск httpd

- Удалён `index.html`
- Создан файл `index.php` с `phpinfo()`;
- Выполнен `chown -R apache:apache /var/www`

```
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
mv: cannot stat '/etc/resolv.conf.bak': No such file or directory
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# cd /var/www/html/www.bahis.net
[root@server.bahis.net www.bahis.net]# ls
index.html
0:17:47 [root@server.bahis.net www.bahis.net]# rm index.html
rm: remove regular file 'index.html'? y
[root@server.bahis.net www.bahis.net]# nano index.php
```

Рисунок 6: Создание `index.php`, настройка прав доступа и перезапуск `httpd`

- Удалён index.html
- Создан файл index.php с phpinfo();
- Выполнен chown -R apache:apache /var/www
- Восстановлены SELinux-контексты

```
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
mv: cannot stat '/etc/resolv.conf.bak': No such file or directory
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# cd /var/www/html/www.bahis.net
[root@server.bahis.net www.bahis.net]# ls
index.html
0:17:47 [root@server.bahis.net www.bahis.net]# rm index.html
rm: remove regular file 'index.html'? y
[root@server.bahis.net www.bahis.net]# nano index.php
```

Рисунок 6: Создание index.php, настройка прав доступа и перезапуск httpd

- Удалён index.html
- Создан файл index.php с phpinfo();
- Выполнен chown -R apache:apache /var/www
- Восстановлены SELinux-контексты
- Перезапущен httpd

```
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# mv /etc/resolv.conf.bak /etc/resolv.conf
mv: cannot stat '/etc/resolv.conf.bak': No such file or directory
[root@server.bahis.net ~]# cat /etc/resolv.conf
nameserver 8.8.8.8
[root@server.bahis.net ~]# cd /var/www/html/www.bahis.net
[root@server.bahis.net www.bahis.net]# ls
index.html
0:17:47 [root@server.bahis.net www.bahis.net]# rm index.html
rm: remove regular file 'index.html'? y
[root@server.bahis.net www.bahis.net]# nano index.php
```

Рисунок 6: Создание index.php, настройка прав доступа и перезапуск httpd

- На клиенте открыт `www.bahis.net`

HTTP Server Test Page powered by `PHP 8.0.30 - phpinfo()`

www.bahis.net

Firefox automatically sends some data to Mozilla so that we can improve your experience. [Choose What I Share](#)

PHP Version 8.0.30



System	Linux serverbahis.net 5.14.0-611.27.1.el9_7.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Feb 6 13:43:43 UTC 2025 x86_64
Build Date	Dec 18 2025 18:46:25
Build System	Rocky Linux release 9.7 (Blue Onyx)
Build Provider	Rocky Enterprise Software Foundation
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-11)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-enchant.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-imagick.ini, /etc/php.d/20-ldap.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mcrypt.ini, /etc/php.d/20-mysqlnd.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-pdo_mysql.ini, /etc/php.d/20-pdo_pgsql.ini, /etc/php.d/20-pdo_sqlite.ini, /etc/php.d/20-posix.ini, /etc/php.d/20-readline.ini, /etc/php.d/20-replicate.ini, /etc/php.d/20-session.ini, /etc/php.d/20-shmop.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sysvmsg.ini, /etc/php.d/20-sysvsem.ini, /etc/php.d/20-sysvshm.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlrpc.ini, /etc/php.d/20-xsl.ini
PHP API	20200930
PHP Extension	20200930
Zend Extension	40200930
Zend Extension Build	API20200930.NTS
PHP Extension Build	API20200930.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled

- На клиенте открыт `www.bahis.net`
- Отображена страница `phpinfo()`

PHP Version 8.0.30	
System	Linux serverbahis.net 5.14.0-611.27.1.el9_7.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Feb 6 13:43:43 UTC 2025 x86_64
Build Date	Dec 18 2025 18:46:25
Build System	Rocky Linux release 9.7 (Blue Onyx)
Build Provider	Rocky Enterprise Software Foundation
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-11)
Architecture	x86_64
Server API	PHP/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-enchant.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-igmp.ini, /etc/php.d/20-intl.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tidy.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlreader.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-zip.ini, /etc/php.d/20-zlib.ini, /etc/php.d/20-zlib.ini
PHP API	20200910
PHP Extension	20200910
Zend Extension	41000910
Zend Extension Build	API20200910.ZTS
PHP Extension Build	API20200910.ZTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled

- На клиенте открыт `www.bahis.net`
- Отобрана страница `phpinfo()`
- Подтверждена интеграция PHP с Apache

PHP Version 8.0.30	
System	Linux serverbahis.net 5.14.0-611.27.1.el9_7.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Feb 6 13:43:43 UTC 2025 x86_64
Build Date	Dec 18 2025 18:00:25
Build System	Rocky Linux release 9.7 (Blue Onyx)
Build Provider	Rocky Enterprise Software Foundation
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-11)
Architecture	x86_64
Server API	PHP/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-enchant.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-igmp.ini, /etc/php.d/20-intl.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mysql.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-sqlite3.ini, /etc/php.d/20-tidy.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlreader.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-zip.ini, /etc/php.d/20-zlib.ini
PHP API	20200910
PHP Extension	20200910
Zend Extension	41000910
Zend Extension Build	API20200910.ZTS
PHP Extension Build	API20200910.ZTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled
Zend Memory Manager	enabled

2.3 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Скопированы конфигурации из `/etc/httpd/conf.d`

```
[root@server.bahis.net www.bahis.net]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.bahis.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.user.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.bahis.net.conf'? y
[root@server.bahis.net www.bahis.net]# p -R /var/www/html/* /vagrant/provision/server/http/var/www/html
bash: p: command not found...
[root@server.bahis.net www.bahis.net]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/index.html'? y
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.bahis.net/index.html'? y
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.bahis.net www.bahis.net]# cp -R /etc/pki/tls/private/www.bahis.net.key /vagrant/provision/server/http/etc/pki/tls/private
```

2.3 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Скопированы конфигурации из `/etc/httpd/conf.d`
- Скопирован веб-контент из `/var/www/html`

```
[root@server.bahis.net www.bahis.net]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.bahis.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.user.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.bahis.net.conf'? y
[root@server.bahis.net www.bahis.net]# p -R /var/www/html/* /vagrant/provision/server/http/var/www/html
bash: p: command not found...
[root@server.bahis.net www.bahis.net]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/index.html'? y
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.bahis.net/index.html'? y
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.bahis.net www.bahis.net]# cp -R /etc/pki/tls/private/www.bahis.net.key /vagrant/provision/ser
ver/http/etc/pki/tls/private
```

2.3 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Скопированы конфигурации из `/etc/httpd/conf.d`
- Скопирован веб-контент из `/var/www/html`
- Созданы каталоги для ключей и сертификатов в `provision`

```
[root@server.bahis.net www.bahis.net]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.bahis.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.user.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.bahis.net.conf'? y
[root@server.bahis.net www.bahis.net]# p -R /var/www/html/* /vagrant/provision/server/http/var/www/html
bash: p: command not found...
[root@server.bahis.net www.bahis.net]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/index.html'? y
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.bahis.net/index.html'? y
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.bahis.net www.bahis.net]# cp -R /etc/pki/tls/private/www.bahis.net.key /vagrant/provision/ser
ver/http/etc/pki/tls/private
```

2.3 Внесение изменений в настройки внутреннего окружения виртуальной машины

- Скопированы конфигурации из `/etc/httpd/conf.d`
- Скопирован веб-контент из `/var/www/html`
- Созданы каталоги для ключей и сертификатов в `provision`
- Помещены файлы `www.bahis.net.key` и `www.bahis.net.crt`

```
[root@server.bahis.net www.bahis.net]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.bahis.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.user.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.bahis.net.conf'? y
[root@server.bahis.net www.bahis.net]# p -R /var/www/html/* /vagrant/provision/server/http/var/www/html
bash: p: command not found...
[root@server.bahis.net www.bahis.net]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/index.html'? y
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.bahis.net/index.html'? y
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.bahis.net www.bahis.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.bahis.net www.bahis.net]# cp -R /etc/pki/tls/private/www.bahis.net.key /vagrant/provision/ser
ver/http/etc/pki/tls/private
```

- В `http.sh` добавлена установка `php`

```
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dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
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Раздел 3

3. Выводы

3.1 Выводы

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- Конфигурация перенесена в provisioning-окружение
- Обеспечена воспроизводимость настройки через `http.sh`